R104/B_depth8 | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 25 | update 36000 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.0914	66.2037	11.416440	11.427978	11.371692	-0.1949	+0.3927	0.4020
TEST ALL	71.3667	66.1802	11.373810	11.385299	11.329932	-0.1966	+0.3866	0.3951

Dataset	n	NSS Top1 (%)	R104 Top1 (%)	NSS cost	R104 cost
TSP TEST	4500	45.888889	47.177778	7.970400719	7.971443854
CVRP TEST	4500	45.977778	48.000000	18.004802368	18.009954948
TSPLIB	49	20.408163	24.489796	8.217907074	8.210711728
CVRPLIB	100	70.000000	62.000000	67.101111717	67.245830364

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 native CE

R104/B_depth8 | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.0914	66.2037	11.416440	11.427978	11.371692	-0.1949	+0.3927	0.4020	per-problem TRAIN SBS
TSP	46.4667	33.1111	8.025333	8.045354	7.990533	-0.2489	+0.4355	0.4237	T2T
CVRP	46.1778	18.7111	18.159697	18.269377	18.064087	-0.6004	+0.5293	0.7027	RELD_CVRP
ATSP	76.8667	47.4889	1.638869	1.676905	1.632471	-2.2682	+0.3919	0.3893	UNICO_MatPOENet
OVRP	41.1556	35.7556	7.461646	7.469579	7.415777	-0.1062	+0.6185	0.6316	MoSES_RF
VRPB	81.0667	80.8444	9.356575	9.354876	9.318750	+0.0182	+0.4059	0.4076	reld_mt
VRPL	37.5111	33.0889	12.431901	12.437959	12.383444	-0.0487	+0.3913	0.3978	MoSES_RF
VRPTW	80.1556	79.9556	18.818368	18.819431	18.766139	-0.0056	+0.2783	0.2941	reld_mt
OVRPTW	83.9111	83.7778	10.142501	10.143730	10.116859	-0.0121	+0.2535	0.2509	reld_mt
OVRPB	88.5778	88.5111	6.356858	6.357254	6.341078	-0.0062	+0.2489	0.2604	reld_mt
OVRPL	41.6444	35.9111	7.438480	7.445709	7.390416	-0.0971	+0.6504	0.6660	MoSES_RF
VRPBL	79.0222	78.5111	9.367325	9.371523	9.328366	-0.0448	+0.4176	0.4199	reld_mt
VRPBTW	78.6667	78.4222	19.944873	19.950430	19.851084	-0.0279	+0.4725	0.4544	reld_mt
VRPLTW	78.8889	78.7556	18.764651	18.765871	18.707843	-0.0065	+0.3037	0.3173	reld_mt
OVRPBL	89.4000	89.2000	6.369207	6.370093	6.352824	-0.0139	+0.2579	0.2697	reld_mt
OVRPBTW	83.6667	83.4444	10.659779	10.661098	10.623245	-0.0124	+0.3439	0.3123	reld_mt
OVRPLTW	83.6889	83.4889	10.156202	10.157423	10.129818	-0.0120	+0.2605	0.2637	reld_mt
VRPBLTW	79.5556	79.4000	19.793468	19.797044	19.706420	-0.0181	+0.4417	0.4304	reld_mt
OVRPBLTW	83.2222	83.2889	10.610189	10.609955	10.571302	+0.0022	+0.3678	0.3444	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R104/B_depth8 | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.3667	66.1802	11.373810	11.385299	11.329932	-0.1966	+0.3866	0.3951	per-problem TRAIN SBS
TSP	47.1778	34.5333	7.971444	7.989621	7.937593	-0.2275	+0.4265	0.4191	T2T
CVRP	48.0000	17.9333	18.009955	18.124983	17.918509	-0.6346	+0.5103	0.6709	RELD_CVRP
ATSP	76.6444	47.2444	1.635607	1.674207	1.629441	-2.3056	+0.3784	0.3777	UNICO_MatPOENet
OVRP	40.5556	33.6444	7.446223	7.455750	7.400487	-0.1278	+0.6180	0.6289	MoSES_RF
VRPB	80.7778	80.4000	9.349189	9.351810	9.315298	-0.0280	+0.3638	0.3666	reld_mt
VRPL	38.0000	32.2667	12.349030	12.356138	12.300958	-0.0575	+0.3908	0.3983	MoSES_RF
VRPTW	81.3778	80.8444	18.688294	18.690545	18.637408	-0.0120	+0.2730	0.2841	reld_mt
OVRPTW	83.8444	83.9333	10.095323	10.093063	10.068135	+0.0224	+0.2700	0.2617	reld_mt
OVRPB	88.2000	88.1333	6.370375	6.370826	6.352225	-0.0071	+0.2857	0.2921	reld_mt
OVRPL	41.5556	34.6000	7.464144	7.472542	7.418033	-0.1124	+0.6216	0.6318	MoSES_RF
VRPBL	79.4889	79.4222	9.394944	9.396379	9.355967	-0.0153	+0.4166	0.4136	reld_mt
VRPBTW	78.8667	78.7556	19.666060	19.665725	19.577666	+0.0017	+0.4515	0.4487	reld_mt
VRPLTW	79.4000	79.2444	18.803048	18.803136	18.745143	-0.0005	+0.3089	0.3280	reld_mt
OVRPBL	89.2000	89.2889	6.375996	6.375741	6.359310	+0.0040	+0.2624	0.2749	reld_mt
OVRPBTW	85.0000	84.8222	10.658432	10.659591	10.624748	-0.0109	+0.3170	0.2898	reld_mt
OVRPLTW	82.9778	82.8667	10.085397	10.086112	10.059001	-0.0071	+0.2624	0.2618	reld_mt
VRPBLTW	79.9333	79.7333	19.702838	19.707068	19.613284	-0.0215	+0.4566	0.4413	reld_mt
OVRPBLTW	83.6000	83.5778	10.662288	10.662149	10.625574	+0.0013	+0.3455	0.3229	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R104/B_depth8 | LIB and locked provenance

Current R104 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	24.4898	10.2041	8.210712	8.240435	8.142167	-0.3607	+0.8418	0.8774
CVRPLIB	62.0000	45.0000	67.245830	67.245364	66.945093	+0.0007	+0.4492	0.5391

SHA256: 532ca192d4c9434c8fc4577598c320f9193a8933e07782b6c82dd549b0e30370

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, complete R104 capacity model; original R88B physical TSP node0 slot3, methodID+34, synchronous two-way interaction, original score MLP, FP32 utility differences/maximin; stable ascending-global-solver-ID ties; original in

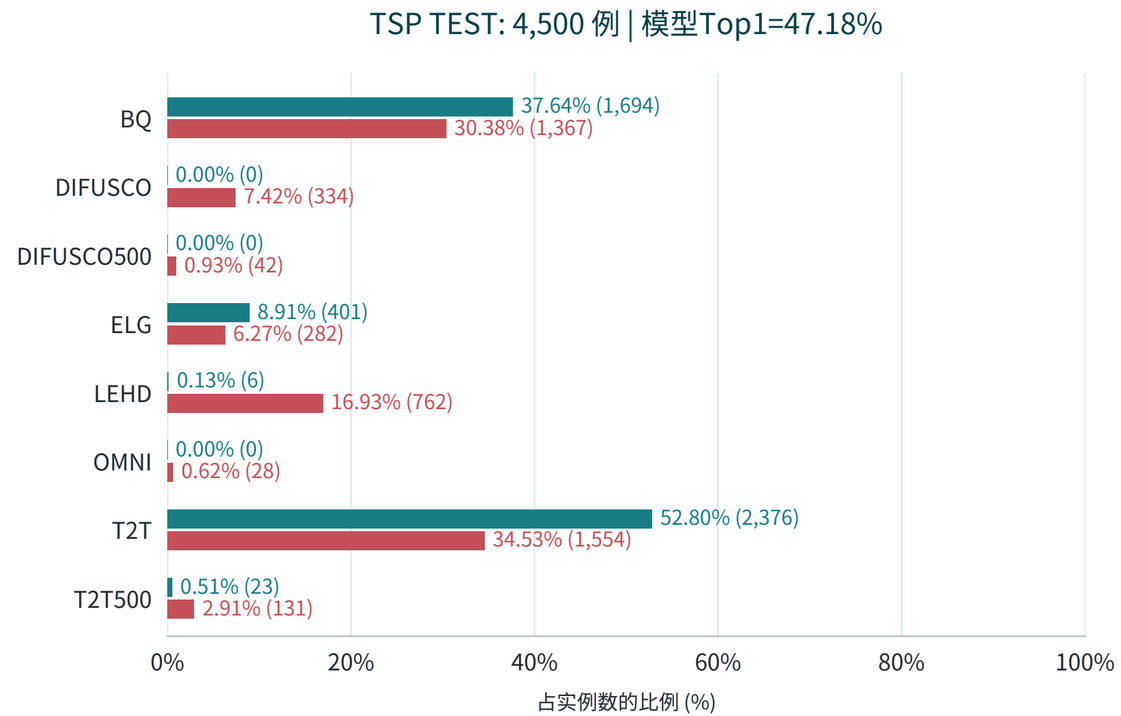
Normalization: new81000 TRAIN only; no TEST/LIB refit.

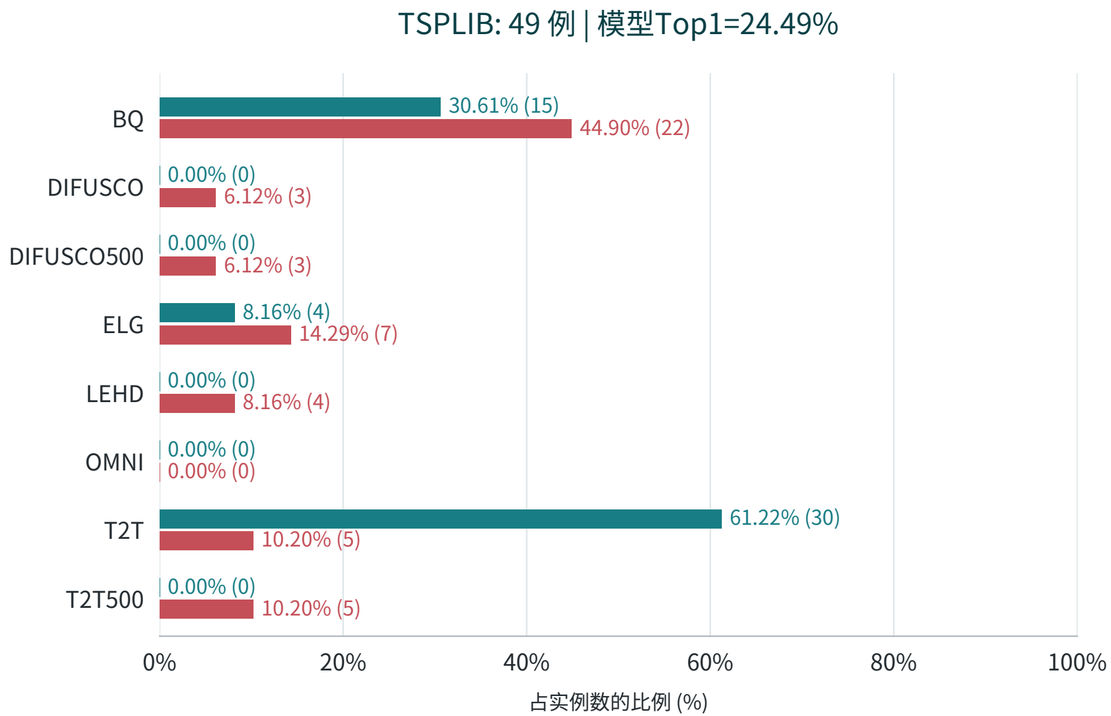
R104/B_depth8 | TSP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

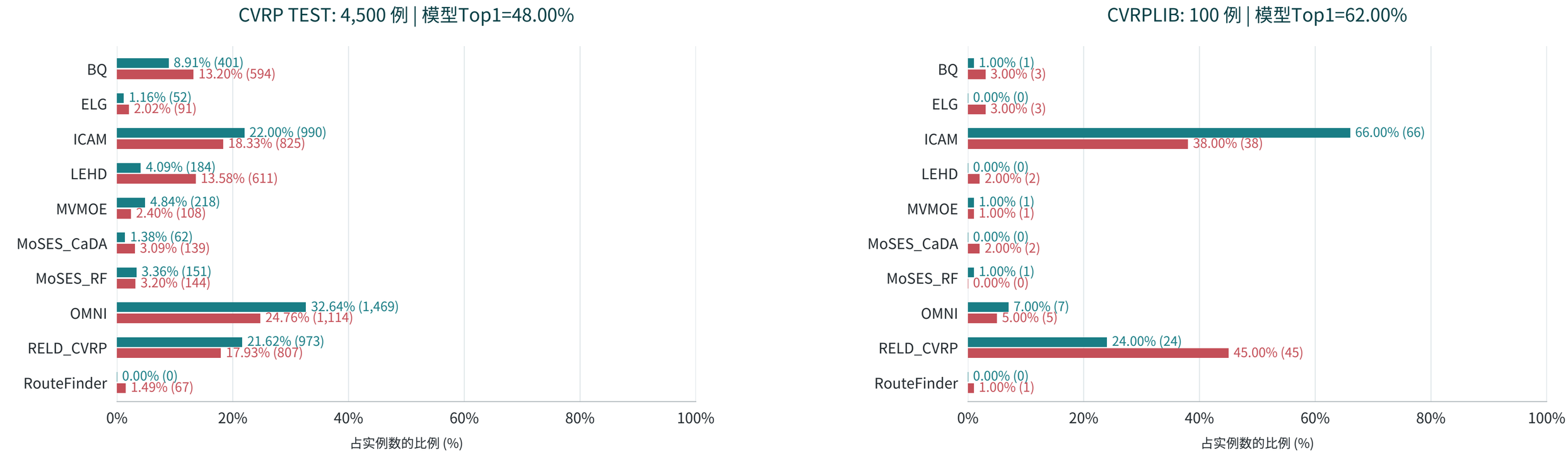
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | CVRP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

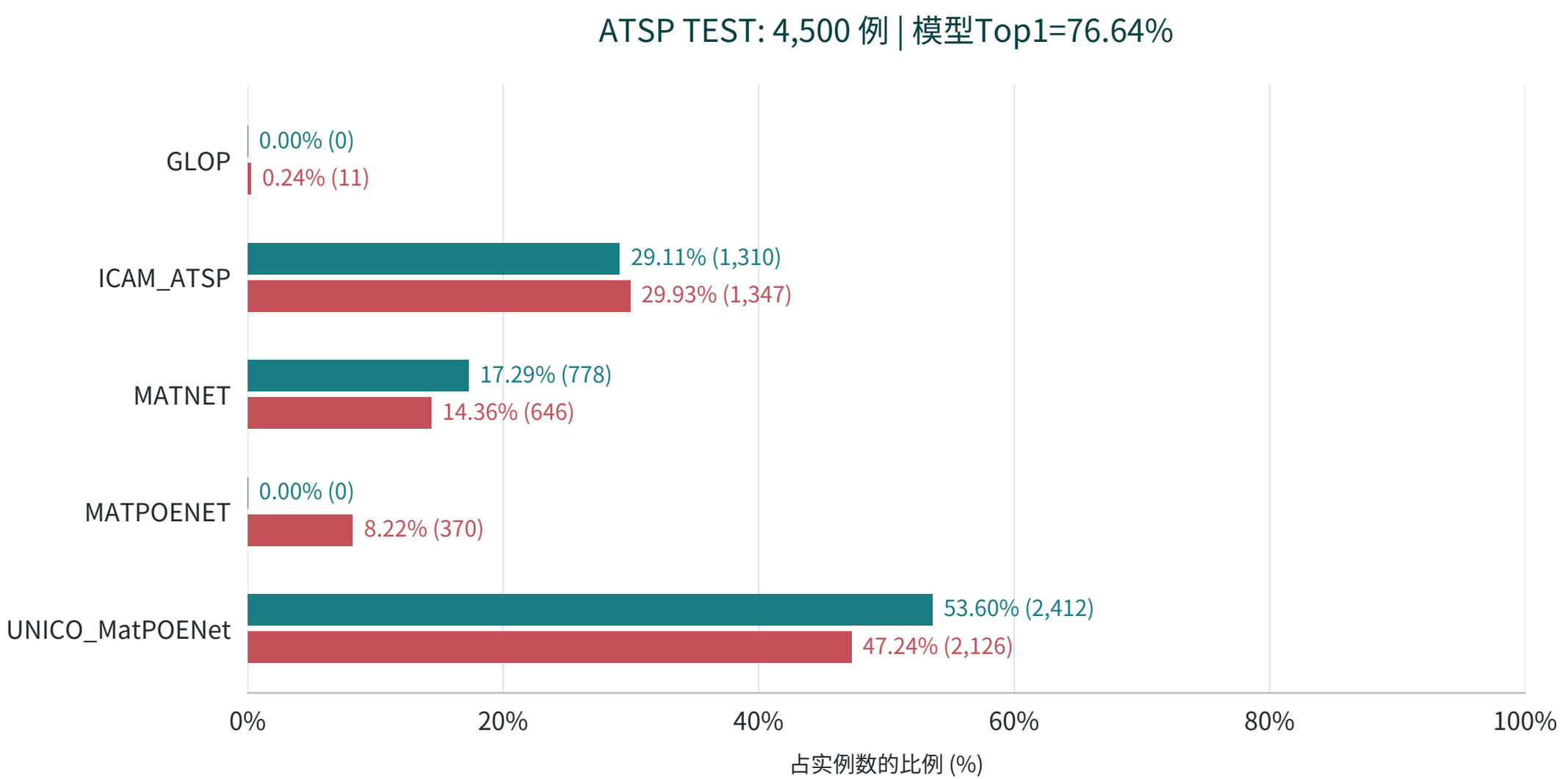
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | ATSP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

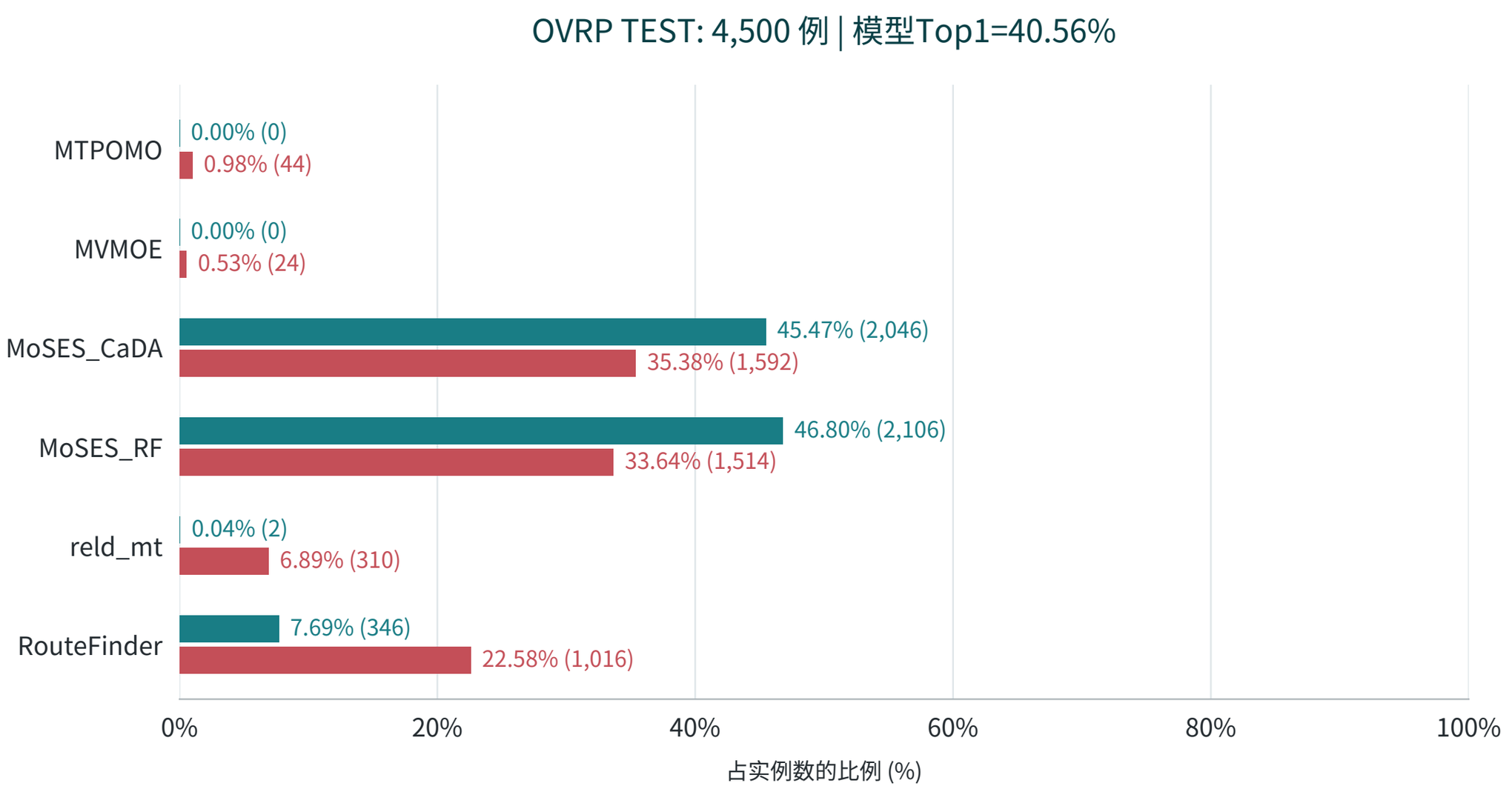
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

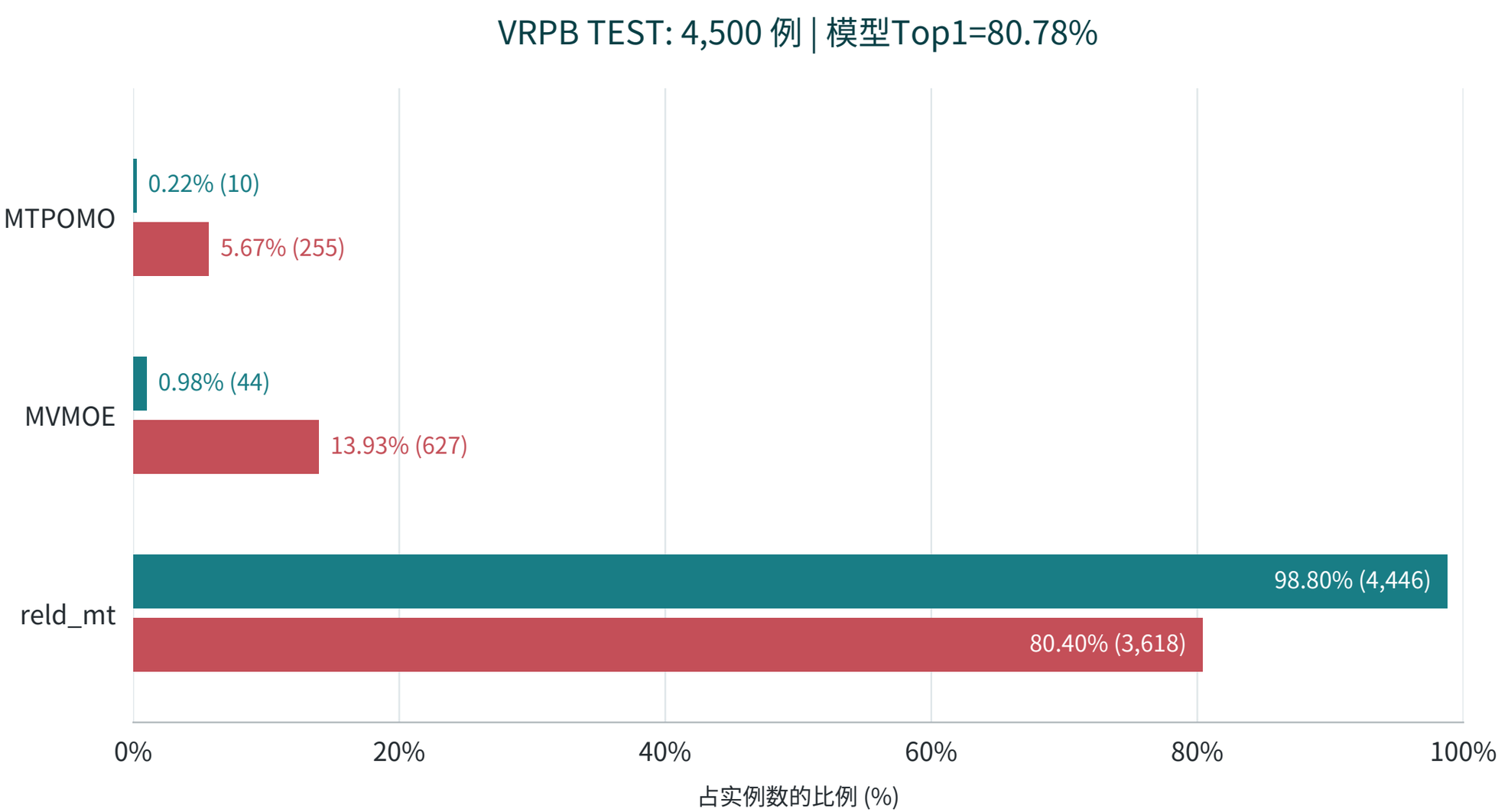
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | VRPB | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

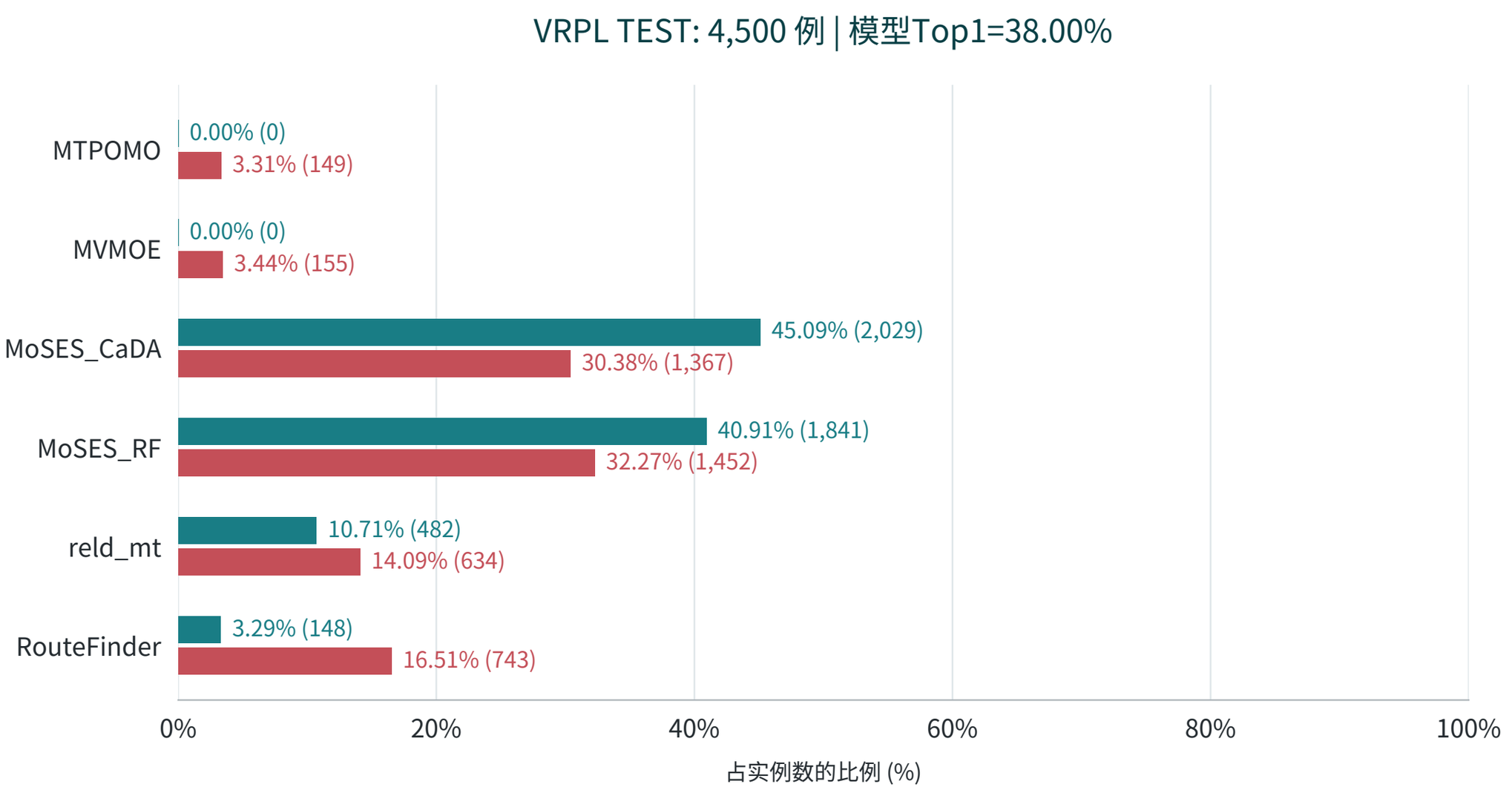
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | VRPL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

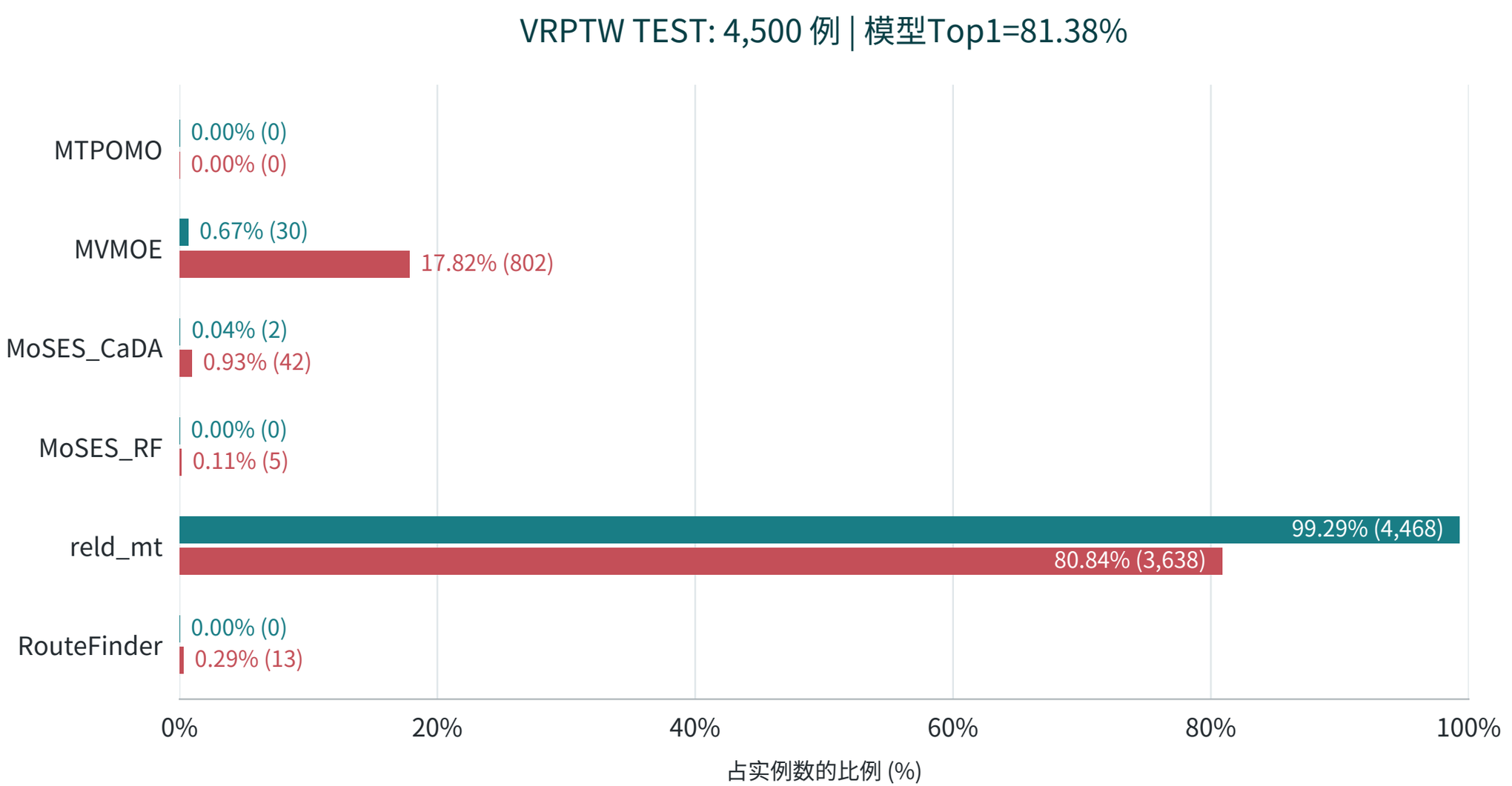
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | VRPTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

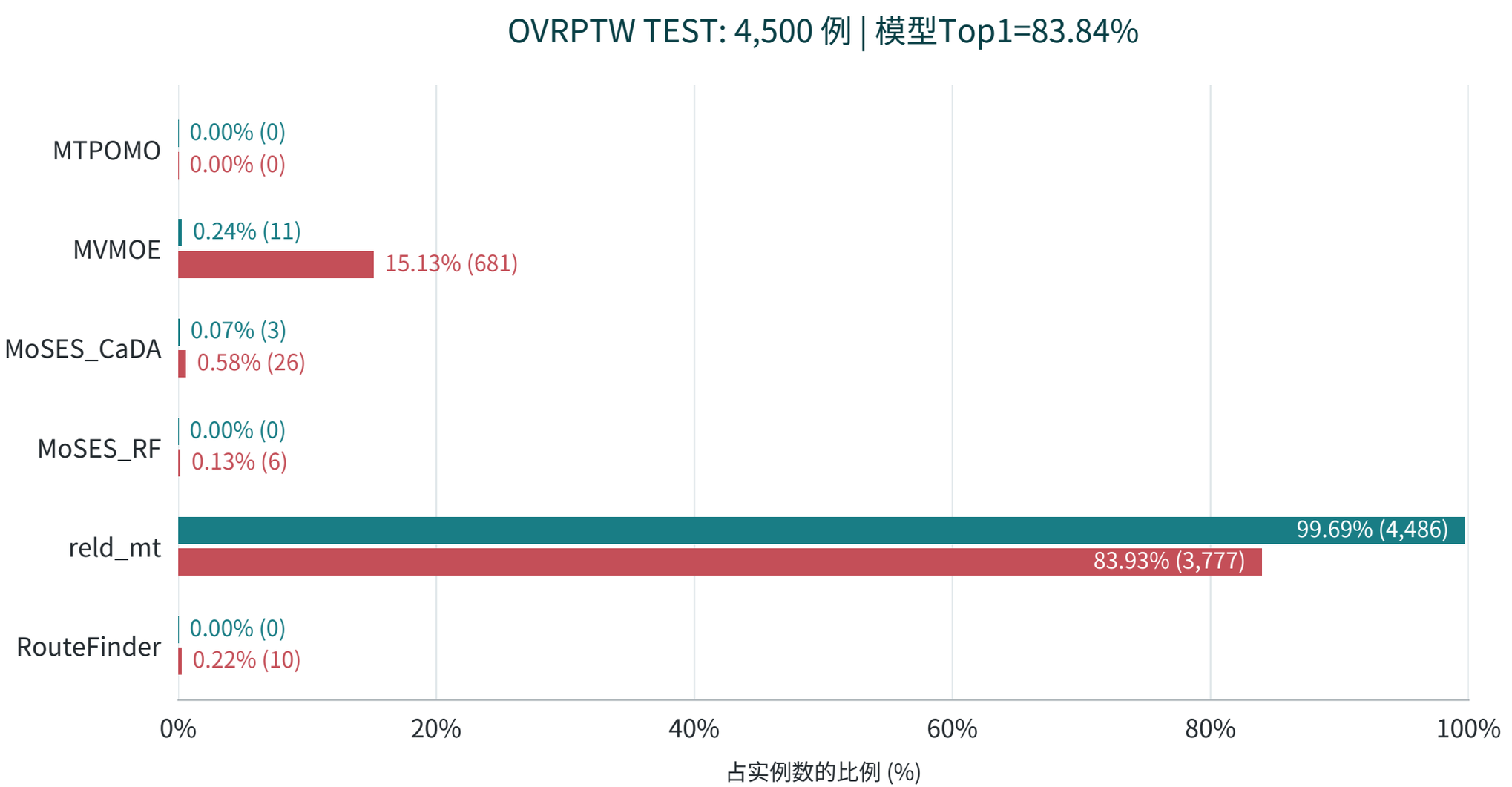
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRPTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

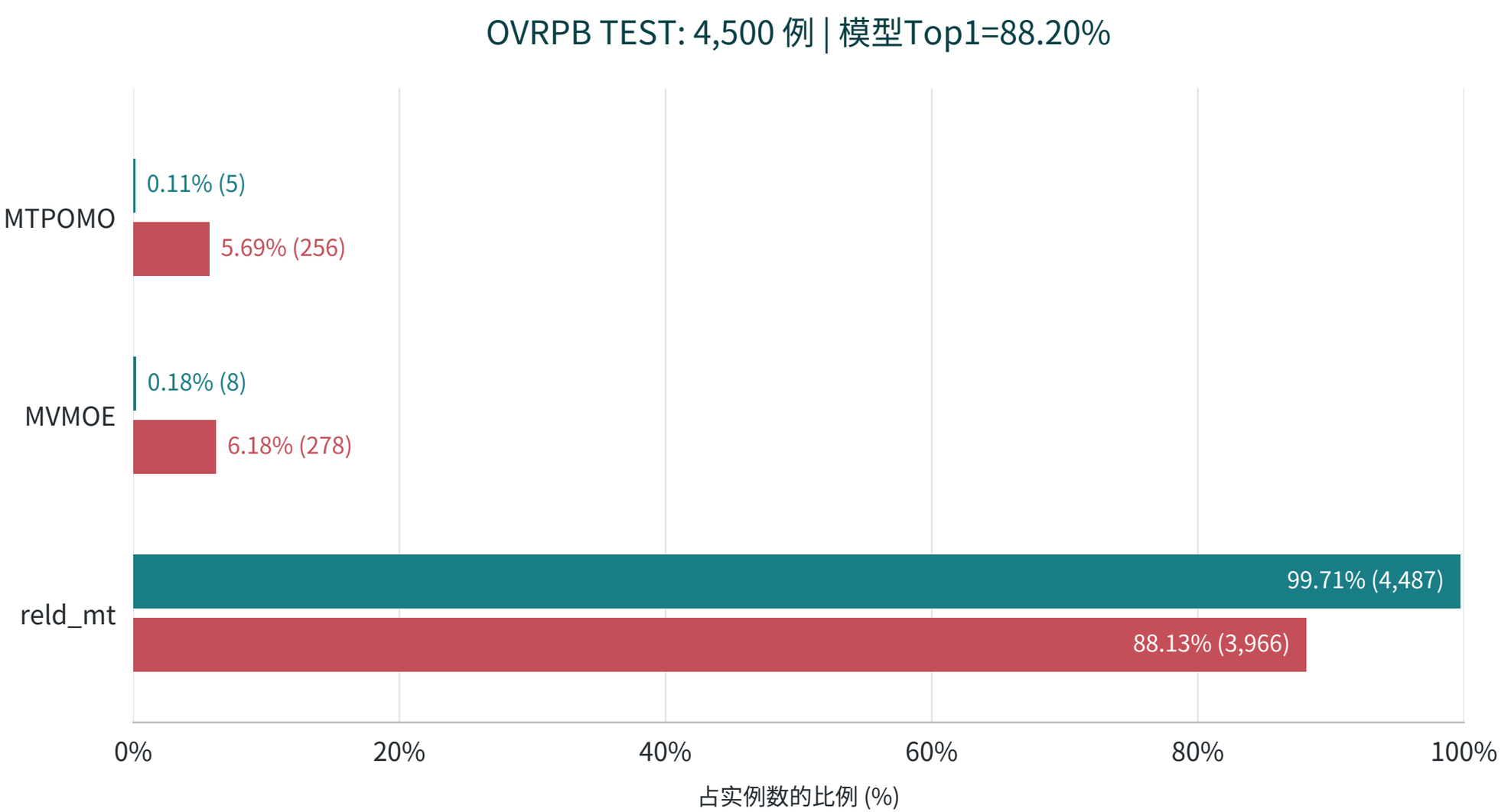
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRPB | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

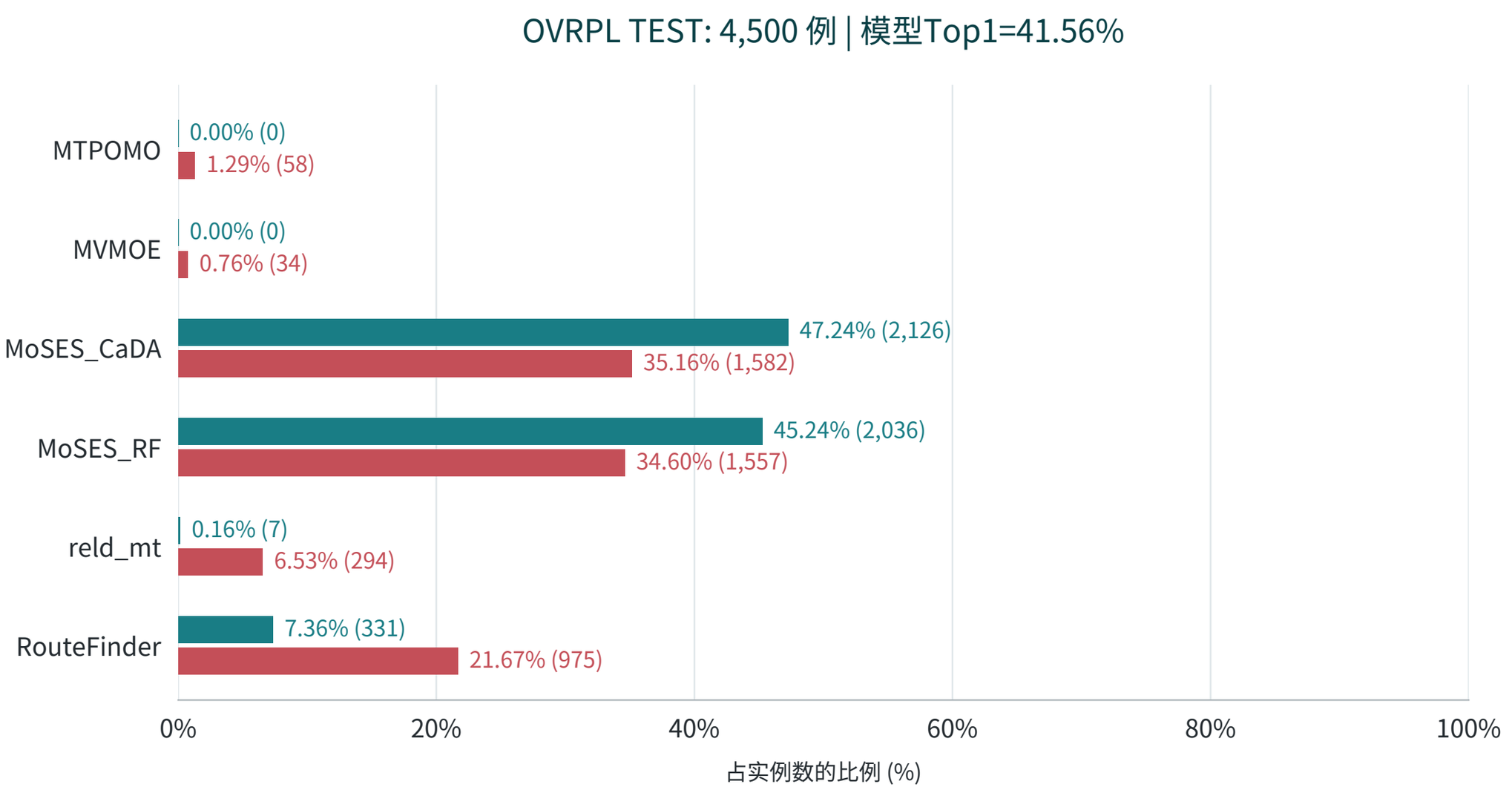
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRPL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

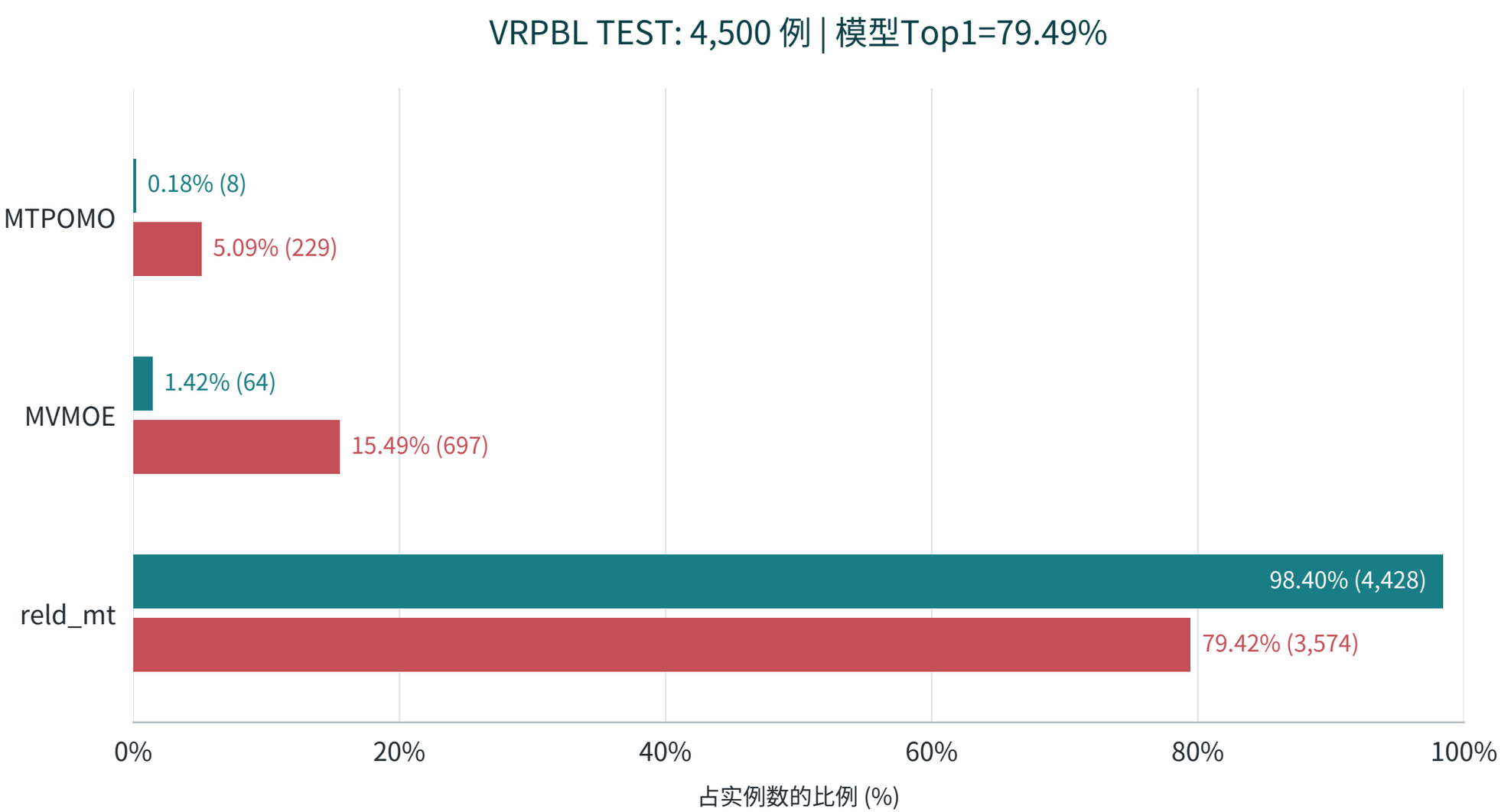
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | VRPBL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

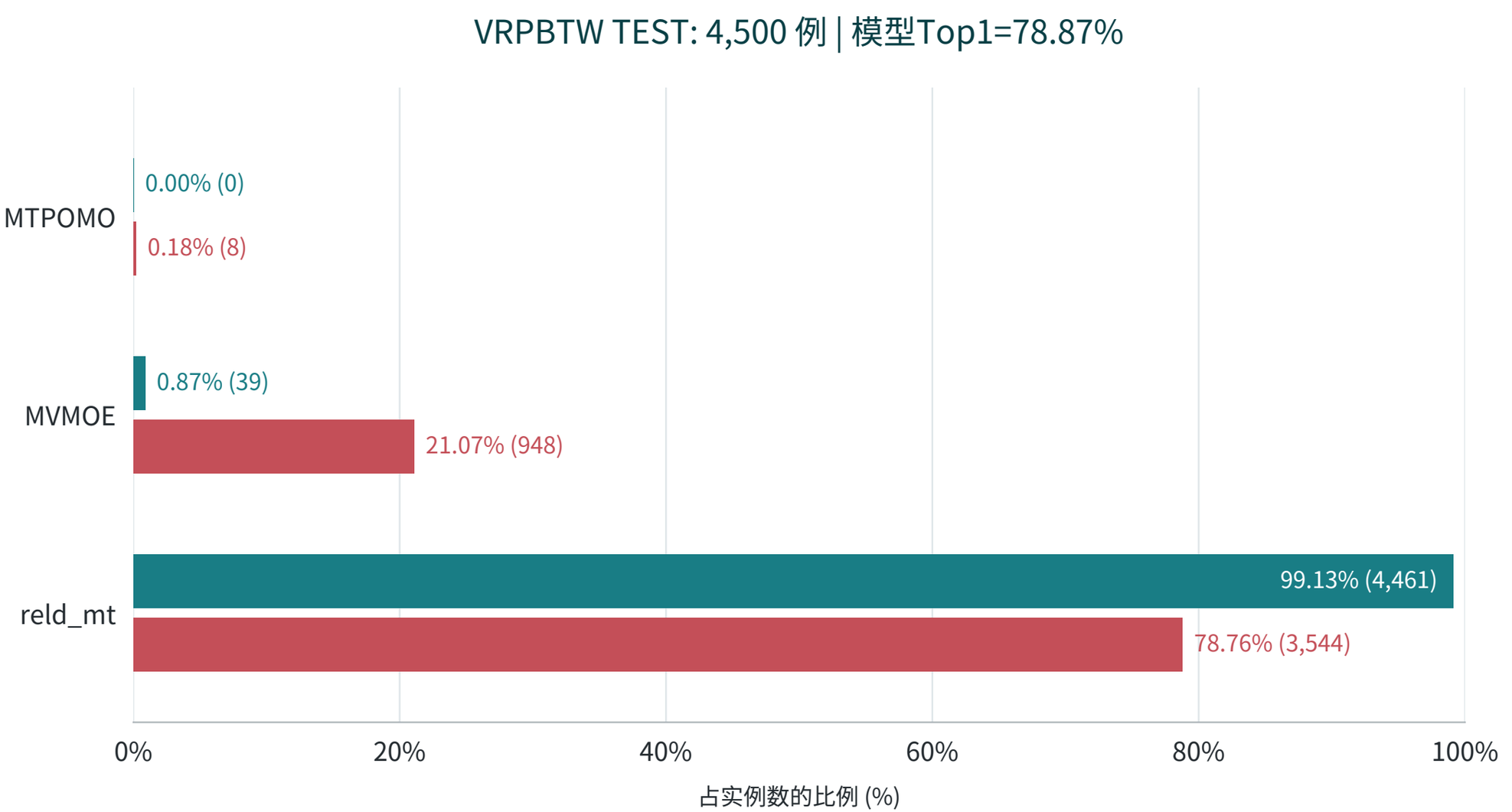
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | VRPBTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

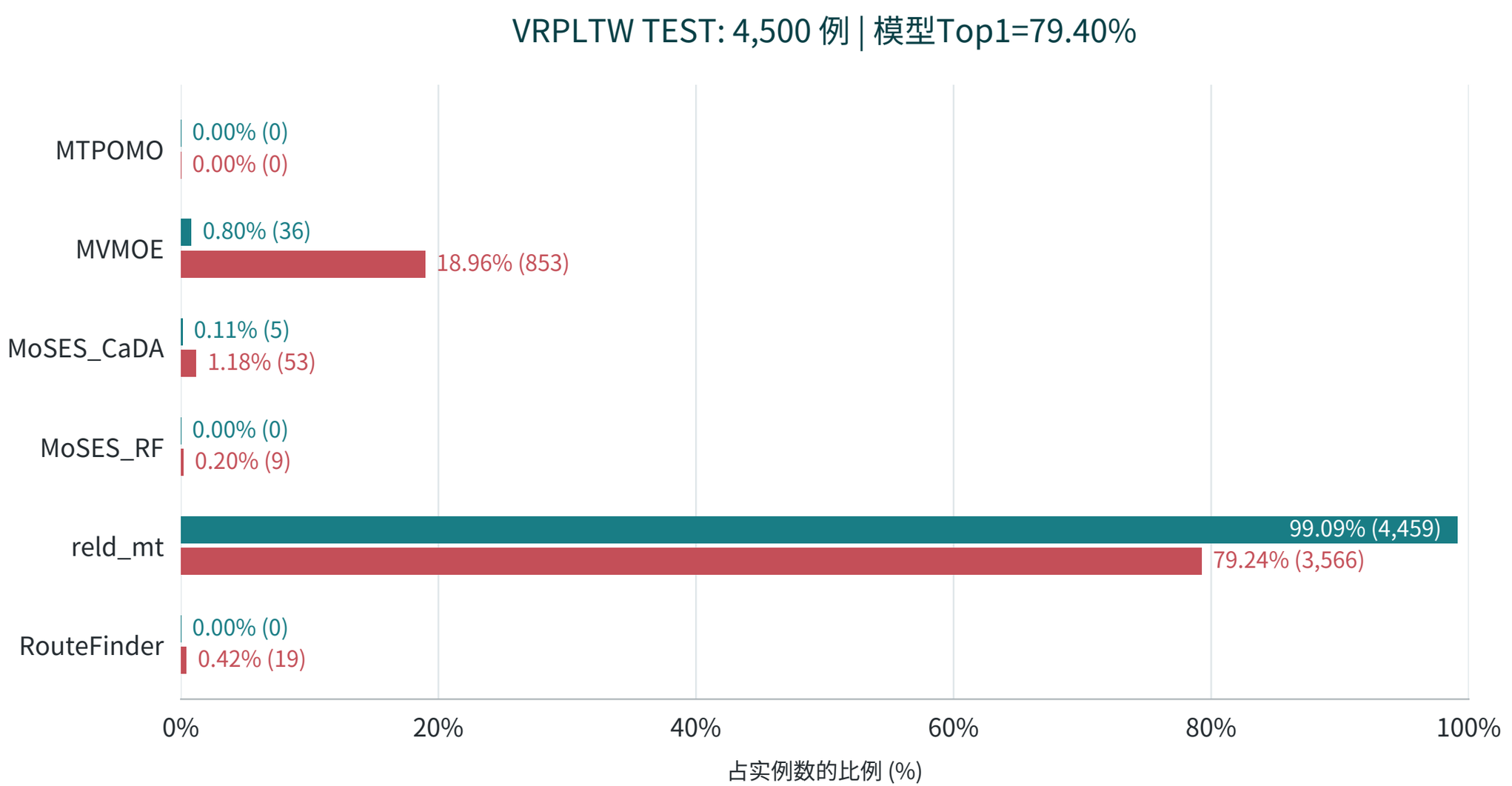
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | VRPLTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

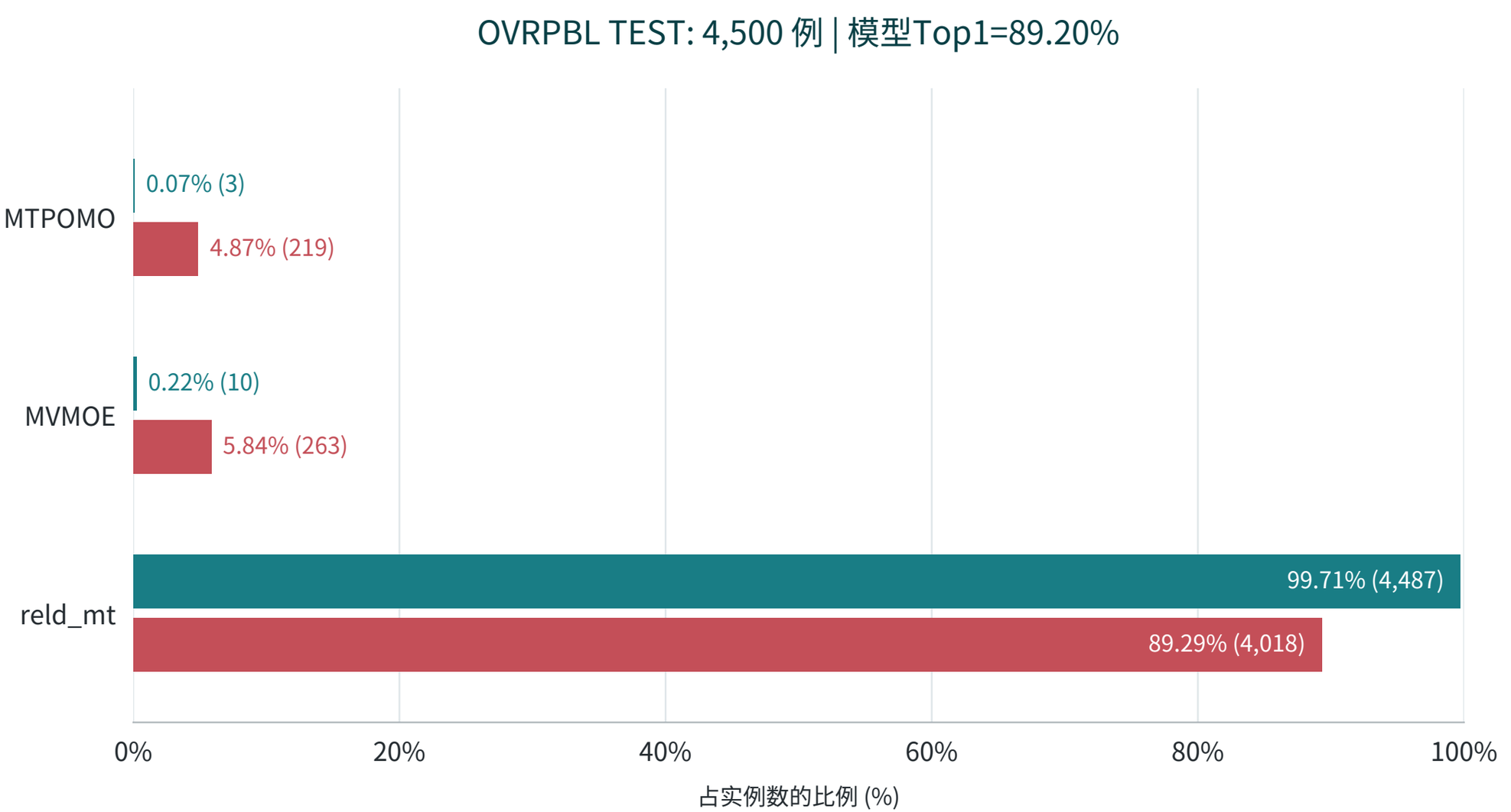
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRPBL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

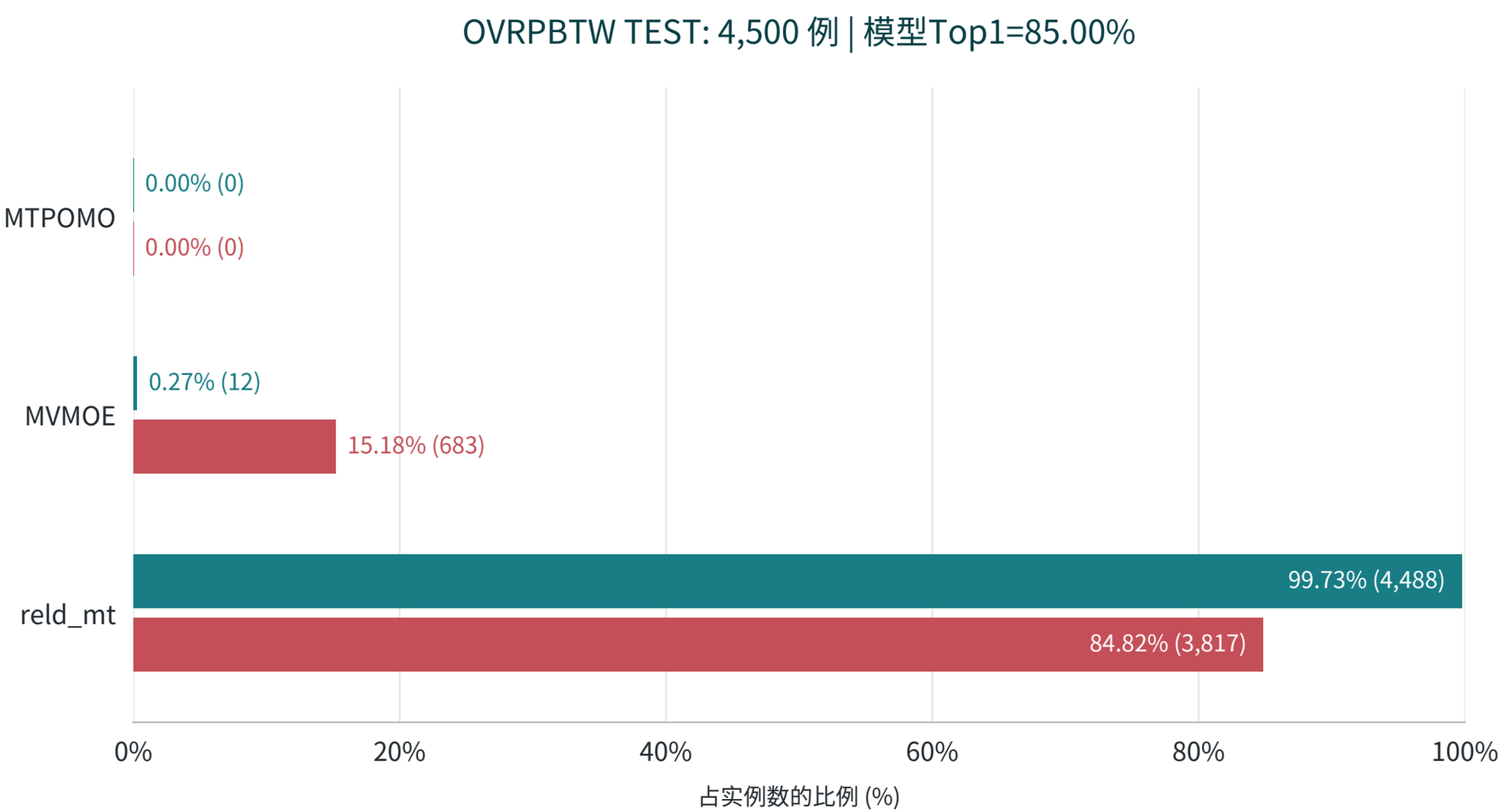
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRPBTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

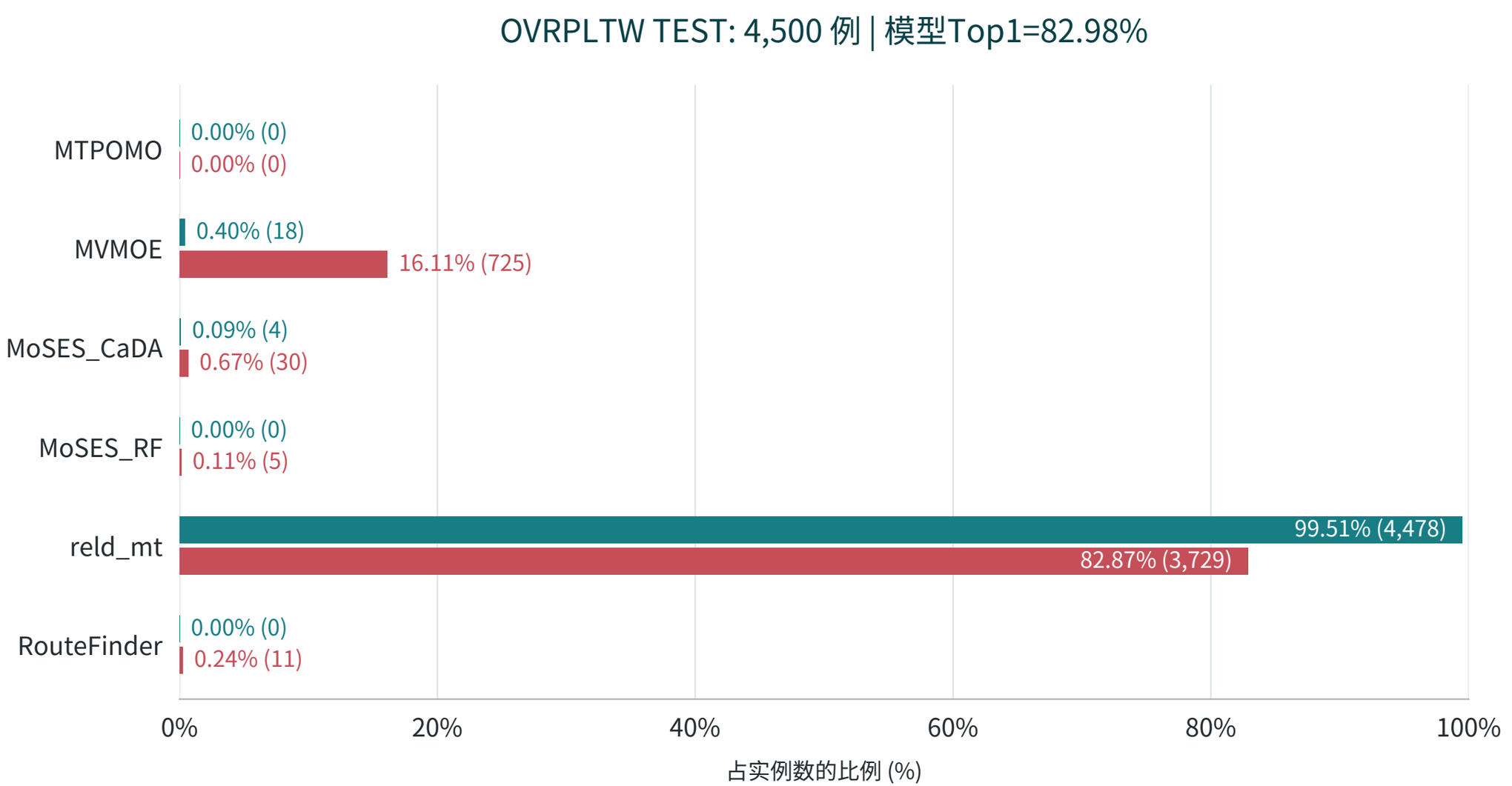
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRPLTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

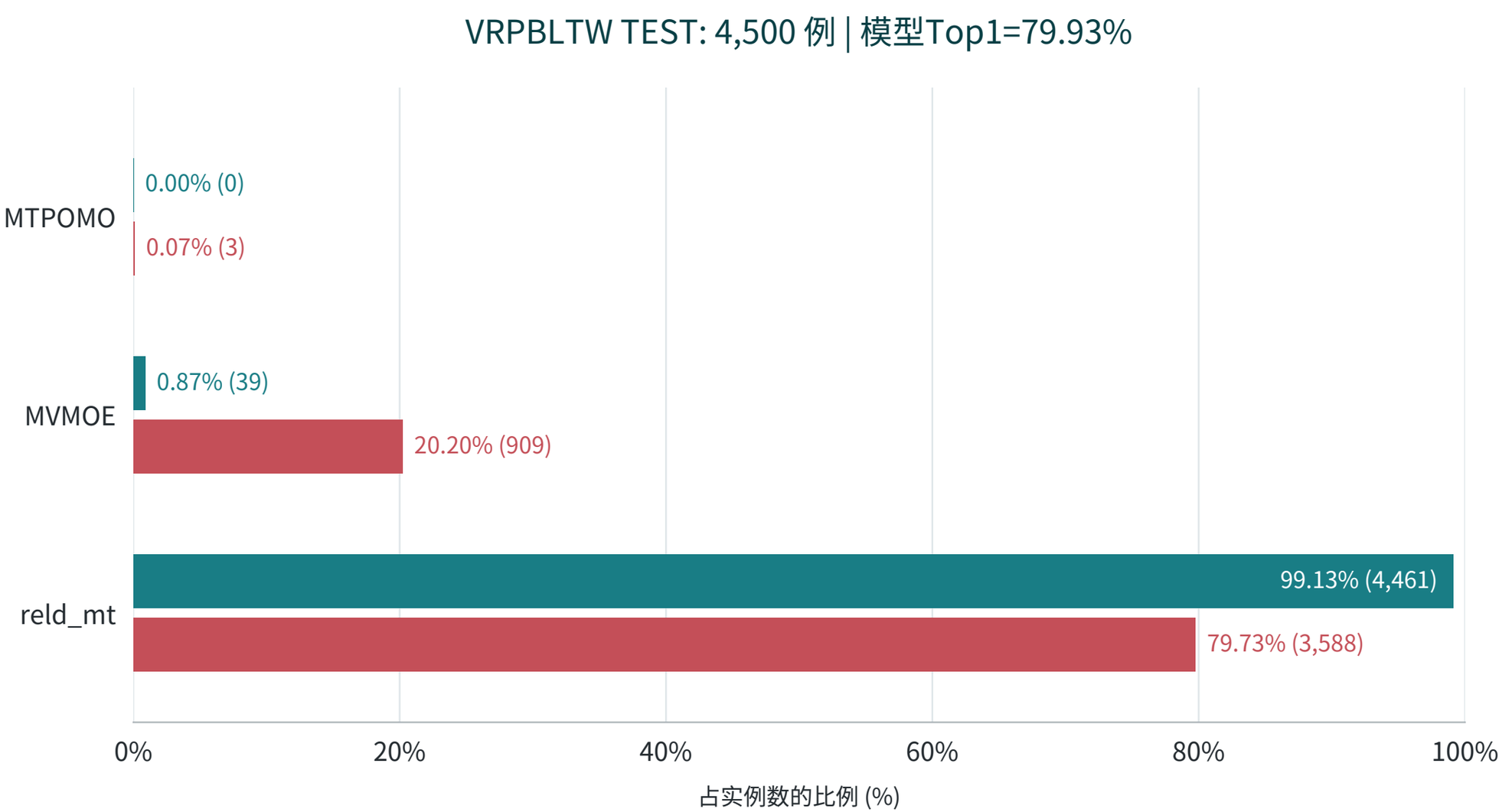
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | VRPBLTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

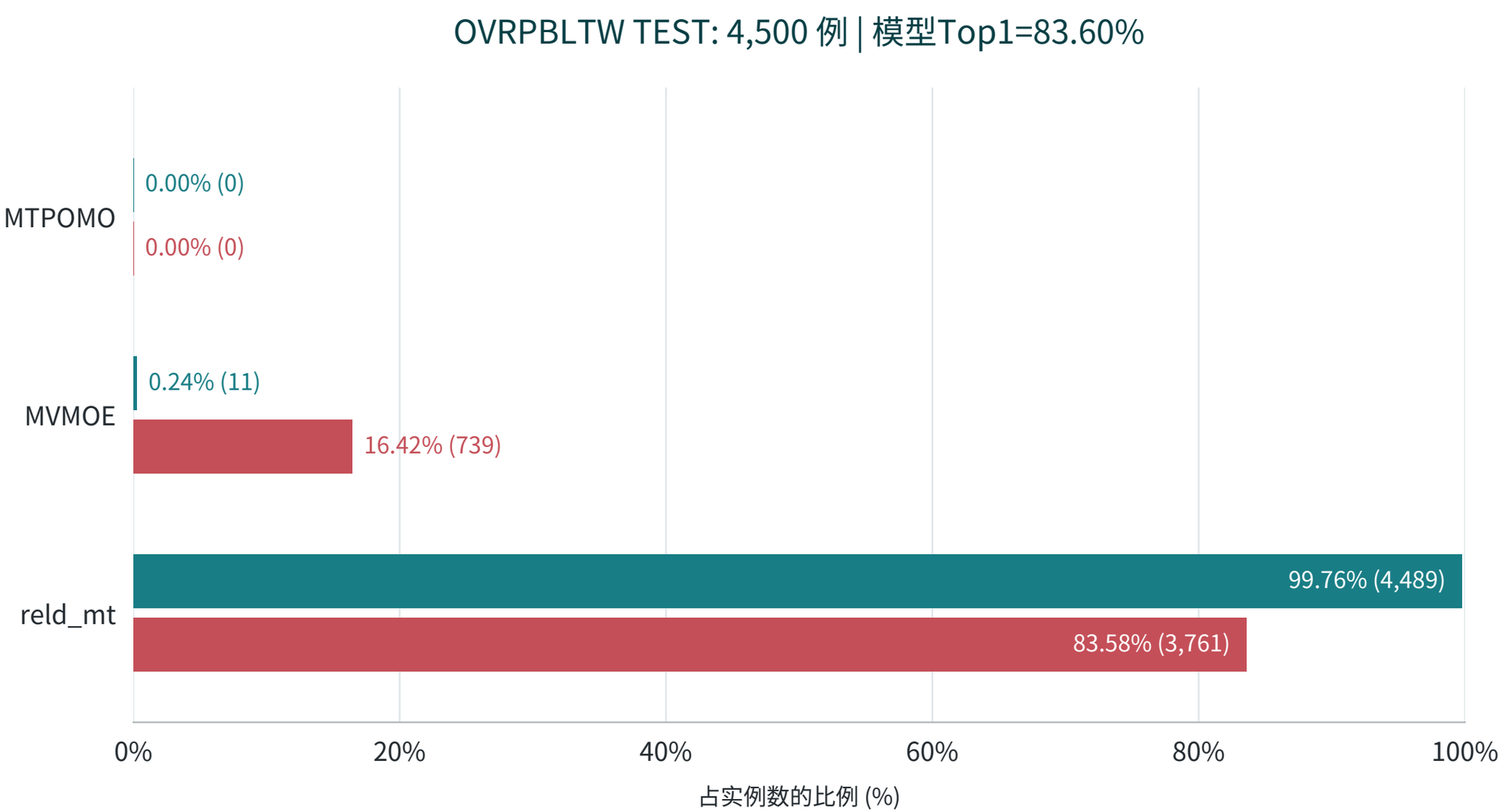
R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.

R104/B_depth8 | OVRPBLTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 532ca192d4c9434c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

R104/B_depth8: stable global solver-ID score ties. Same R104 lock; LIB inference only; no training/reselection/Rxx mixing.